

UST v5 — New Theorems T29–T32

Compact Object Spectrum, Solar Fusion,
Universal Time Unit, and Open Questions

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T29 — Compact Object Spectrum

$$\frac{r_{Om}^{wh}}{r_{Om}^{bh}} = \frac{N_b}{C_{cb}} = 1.728845 \approx \sqrt{3} \quad \Delta\% = 0.185 \quad (1)$$

T30 — Solar Fusion Tunnelling

$$\eta = 6\pi \cdot N_b \cdot C_{cb} = 4.376223 \quad \text{observed: } 4.383722 \quad \Delta\% = 0.171 \quad (2)$$

T32 — Universal Time Unit

$$1 \text{ UST-s} = N_b \times 1 \text{ SI-s} \quad \kappa \cdot dt_{\text{UST}} = \pi \text{ (exact)} \quad (3)$$

$$t_H \times N_b = 9.191984 \text{ Gyr} \approx \nu_{\text{CS-133}} = 9.192632 \text{ GHz} \quad \Delta\% = 0.007 \quad (4)$$

Open Question — A_s and RA^2

An observed numerical relationship:

$$A_s \approx 24 \times RA^2 = 24 \times (N_b - N_m)^2 = 2.1116 \times 10^{-9} \quad (5)$$

Planck 2018: $A_s = 2.099 \times 10^{-9}$, $\Delta\% = 0.60$.

The factor 24 lacks a first-principles derivation. This remains an **open question** pending CMB-S4 data (2028).

Summary

Theorem	Content	Predicted	Observed	$\Delta\%$
T29	Compact spectrum $\approx \sqrt{3}$	1.728845	1.732051	0.185
T30	Solar $\eta = 6\pi N_b C_{C_b}$	4.376223	4.383722	0.171
T32a	$\kappa \cdot dt = \pi$	exact	exact	0.000
T32b	$t_H \times N_b \approx \nu_{C_s}$	9.1920 Gyr	9.1926 GHz	0.007
$A_s?$	$24 \times RA^2$ (open)	2.112×10^{-9}	2.099×10^{-9}	0.60